TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 10

a. NaCl does not contain any hydrogen atom to which diffuse and polarized functions could be assigned.

b. Charge values for C in CH_4:

Basis set	Mulliken	Fit to ESP	AIM
6-31++G(d)	-1.00	-0.46	-0.27
6-31++G(d,p)	-0.79	-0.44	-0.18

Charge values for Na in NaCl:

Basis set \ Method	Mulliken	Fit to ESP	AIM
6-31++G(d)	0.56	0.76	0.88

The charges assigned to the atoms by the three methods differ significantly from each other. This is due to the completely different procedures used. The molecule NaCl is less ionic than bulk NaCl.

c. 6-31++G(d,p) has three basis functions more on each H atom than 6-31++G(d): p_x, p_y and p_z. Since more AOs are available on H centers, they are able to "acquire" more electronic charge. The negative charge of the C atom decreases accordingly.

The basis set affects the spatial distribution of electron density and thus the charges on the atoms. The Mulliken population analysis is strongly basis set dependent because the charges of the atoms are not defined in a spatial sense, but via the basis functions centered at that atoms. Since AOs are not spatially confined their tails fill also regions in the neighborhoods of other atomic centers which leads to the observed strong basis set dependence.

d. The Mulliken population analysis should be used only to compare similar molecules and for investigation of trends. It is usually quite dangerous to take the charge values in in absolute sense, in particular since they strongly depend on the choice of basis set.

e. The dipole moments are:

Method	LSDA	Mulliken	ESP	AIM	exp.[a]
Dipole moment [Debye]	8.8	6.4	8.7	10.0	8.5 ± 0.2

[a]P. A. Tate, M. W. Strandberg, *J. Chem.Phys.* **1954**, *22*, 1380.

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